

SmolVLM

Redefining Small and Efficient Multimodal Models

Authors: Hugging Face & Stanford

Enabling resource-efficient inference on mobile and edge devices.

256M

PARAMETERS

< 1GB

GPU RAM

Large VLMs Are Powerful but Impractical for Edge Deployment

State-of-the-art Vision-Language Models (VLMs) require massive compute and memory, heavily restricting their use on mobile and edge devices.

- **High Memory Ceiling:** Models like Molmo-A1B-7B require nearly 28GB of RAM.
- **Cloud Dependency:** Large models force data to the cloud, introducing latency and privacy risks.
- **The Goal:** Bring multimodal reasoning fully on-device without compromising accuracy.

The Efficiency Gap

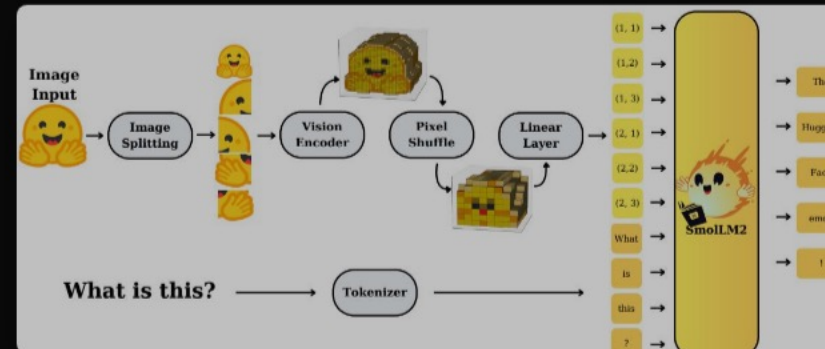
27.7 GB

0.8 GB

SmolVLM Architecture

Image Splitting, Pixel Shuffle, and SmolLM2 Backbone

- **1. Image Splitting:** High-res images split into sub-images to preserve detail.
- **2. Vision Encoder:** Processed through a SigLIP vision encoder.
- **3. Pixel Shuffle:** Compresses spatial features into channels (space-to-depth), drastically reducing token count.
- **4. Projection & LLM:** Linear projection into the SmolLM2 language model's input space.



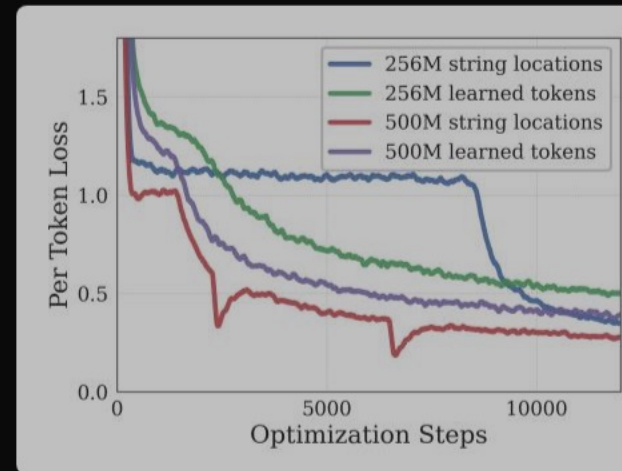
Smaller Vision Encoders Outperform Larger Ones in Compact VLMs

Finding 1: Balanced encoder-LM parameter allocation matters. Counter-intuitively, smaller vision encoders yield better results when paired with highly constrained language models.

Aggressive Pixel Shuffle ($r=4$) Cuts Visual Tokens 16×

Finding 3: Pixel shuffle rearranges spatial features into channels, easing attention overhead and improving long-context modeling.

- **Token Reduction:** Reduces visual tokens by a factor of r^2 .
- **Small VLM Sweet Spot:** While larger models typically use $r=2$ (4× reduction), highly compact VLMs perform significantly better with $r=4$ (16× reduction).
- **Efficiency:** Frees up the context window for complex reasoning and multi-image inputs.



Extending Context from 2K to 16K Tokens

Finding 2: Performance improves significantly with extended context lengths, unlocking higher image resolutions and video support.

- **RoPE Base Adjustment**

Increased from 10K to 273K to support exponentially longer sequences.

- **Mixed-Context Fine-Tuning**

Trained on a curated mix of short and long-context data to maintain accuracy across lengths.

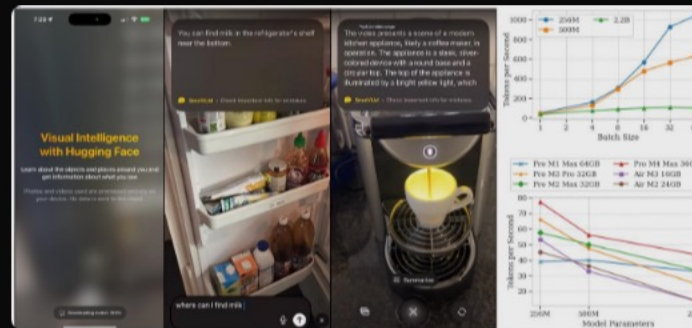
- **Model Tiers**

SmolVLM-2.2B utilizes a full **16K context** window.

Smaller variants (256M, 500M) efficiently operate at **8K context**.

Real-Time Inference: Up to 80 Tokens/Second on Device

Processed entirely locally via the HuggingSnap iOS app with zero cloud dependency.



- **Pro M1 Max 64GB:** Achieves ~80 tokens/sec with SmoIVLM-256M.
- **Batch Scaling:** At batch size 64, throughput reaches nearly 1000 tokens/sec on high-end hardware.
- **MacBook Air M3 16GB:** Smooth processing at ~55 tokens/sec for the 256M model.
- **Linear Scaling:** Throughput scales roughly linearly with batch size for the smallest models.

Key Takeaways

Design Principles for Efficient Multimodal Models

- **1. Balance Compute**
Smaller vision encoders outperform massive ones when paired with compact LMs.
- **2. Aggressive Compression**
Pixel shuffle ($r=4$) crucially reduces token load by 16× for small VLMs.
- **3. Extend Context**
Scaling context to 8K/16K tokens directly unlocks high-res and video capabilities.
- **4. Sub-1GB Reality**
High-quality multimodal inference is actively viable on standard smartphones.
- **5. Fully Open Source**
Weights, datasets, and the HuggingSnap mobile app are entirely open.